# Yufei Zhang


CONTACT INFORMATION

**Office:** 803, Weeks Building, 16-18 Prince's Gardens,
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RESEARCH INTERESTS

Stochastic Control and Games, Mathematical and Computational Finance, Theory and Applications of Machine Learning.

ACADEMIC APPOINTMENTS

**Imperial College London**, United Kingdom
Department of Mathematics

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|---|---|
| Associate Professor (Reader) in Mathematical Finance and Machine Learning | Sep. 2026-present |
| Associate Professor (Senior Lecturer) in Mathematical Finance and Machine Learning | Sep. 2023-Aug. 2026 |
| Director of the MSc in Mathematics and Finance | Sep. 2025-Aug. 2026 |

**London School of Economics**, United Kingdom

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| Assistant Professor at Department of Statistics | Sep. 2021-Aug. 2023 |

EDUCATION

**University of Oxford**, United Kingdom

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|---|---|
| D.Phil., Mathematics | Oct. 2017-June 2021 |

**The Chinese University of Hong Kong**, Hong Kong

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| M.Phil., Mathematics | Aug. 2015-July 2017 |
| M.Sc., Mathematics | Aug. 2013-June 2015 |
| B.B.A., Insurance, Financial and Actuarial Analysis | Aug. 2008-June 2013 |

AWARDS

- JPMorganChase Faculty Research Award, *JPMorganChase*, 2025.
- Simons Foundation Fellowship, *Isaac Newton Institute for Mathematical Sciences*, 2025.
- The Mathematical Institute DPhil Thesis Prize, *University of Oxford*, 2021.
- G-Research PhD Prize in Maths and Data Science, *G-Research*, 2020.
- Academic Support Grant, *The Queen's College, University of Oxford*, 2017.
- Departmental Studentship, *Mathematical Institute, University of Oxford*, 2017–2021.

GRANTS

- **JPMorganChase**, Solo PI, *Decentralized Multi-agent Learning in Finance: From Theory to Market Implications*, $50,000, Oct. 2025-Oct. 2026.
- **Nomura International Plc**, Co-I, *Computational Methods for Counterparty Risk*, £190,000, Oct. 2025-Feb. 2029.
- **esure Services Limited and Accenture (UK) Limited**, PI, *Reinforcement Learning for Insurance Pricing*, £95,000, Nov. 2022- Apr. 2023.
- **Isaac Newton Institute**, Program Proposer, *Bridging Stochastic Control And Reinforcement Learning*, £119,840, Nov. 3-28, 2025.

JOURNAL PUBLICATIONS

[1] Eyal Neuman and Yufei Zhang, *Statistical learning with sublinear regret of propagator models*, The Annals of Applied Probability, 36 (2026), pp. 2873-2915.

[2] Lukasz Szpruch, Tanut Treetanthiploet, and Yufei Zhang, *Exploration-exploitation trade-off for continuous-time episodic reinforcement learning with linear-convex models*, The Annals of Applied Probability, 36 (2026), pp. 959–1001.

[3] Yanwei Jia, Du Ouyang, and Yufei Zhang, *Accuracy of discretely sampled stochastic policies in continuous-time reinforcement learning*, SIAM Journal on Control and Optimization, 64 (2026), pp. 1889–1929.

[4] Bekzhan Kerimkulov, James-Michael Leahy, David Šiška, Łukasz Szpruch, and Yufei Zhang, *A Fisher-Rao gradient flow for entropy-regularised Markov decision processes in Polish spaces*, Foundations of Computational Mathematics, 26 (2026), pp. 2395–2469.

[5] Tanut Treetanthiploet, Łukasz Szpruch, and Yufei Zhang, *$\epsilon$-policy gradient for online pricing*, Applied Mathematics and Optimization, 94 (2026), pp. 24.

[6] Bekzhan Kerimkulov, David Šiška, Łukasz Szpruch, and Yufei Zhang, *Mirror descent for stochastic control problems with measure-valued controls*, Stochastic Processes and Their Applications, 190 (2025), pp. 104765.

[7] Deven Sethi, David Šiška, and Yufei Zhang, *Entropy annealing for policy mirror descent in continuous time and space*, SIAM Journal on Control and Optimization, 63 (2025), pp. 3006–3041.

[8] Xin Guo, Xinyu Li, and Yufei Zhang, *An $\alpha$-potential game framework for $N$-player games*, SIAM Journal on Control and Optimization, 63 (2025), pp. 2964–3005.

[9] Xin Guo and Yufei Zhang, *Towards an analytical framework for dynamic potential games*, SIAM Journal on Control and Optimization, 63 (2025), pp. 1213–1242.

[10] Christoph Reisinger, Wolfgang Stockinger, and Yufei Zhang, *A fast iterative PDE-based algorithm for feedback controls of nonsmooth mean-field control problems*, SIAM Journal on Scientific Computing, 46 (2024), pp. A2737–A2773.

[11] Michael Giegrich, Christoph Reisinger, and Yufei Zhang, *Convergence of policy gradient methods for finite-horizon exploratory linear-quadratic control problems*, SIAM Journal on Control and Optimization, 62 (2024), pp. 1060–1092.

[12] Lukasz Szpruch, Tanut Treetanthiploet, and Yufei Zhang, *Optimal scheduling of entropy regulariser for continuous-time linear-quadratic reinforcement learning*, SIAM Journal on Control and Optimization, 62 (2024), pp. 135–166.

[13] Christoph Reisinger, Wolfgang Stockinger, and Yufei Zhang, *Linear convergence of a policy gradient method for some finite horizon continuous time control problems*, SIAM Journal on Control and Optimization, 61 (2023), pp. 3526–3558.

[14] Christoph Reisinger, Wolfgang Stockinger, and Yufei Zhang, *A posteriori error estimates for fully coupled McKean-Vlasov forward-backward SDEs*, IMA Journal of Numerical Analysis, 44 (2024), pp. 2323–2369.

[15] Xin Guo, Anran Hu and Yufei Zhang, *Reinforcement learning for linear-convex models with jumps via stability analysis of feedback controls*, SIAM Journal on Control and Optimization, 61 (2023), pp. 755–787.

[16] Matteo Basei, Xin Guo, Anran Hu and Yufei Zhang, *Logarithmic regret for episodic continuous-time linear-quadratic reinforcement learning over a finite-time horizon*, Journal of Machine Learning Research, 23 (2022), pp. 1–34.

[17] Christoph Reisinger and Yufei Zhang, *Regularity and stability of feedback relaxed controls*, SIAM Journal on Control and Optimization, 59 (2021), pp. 3118–3151.

[18] Kazufumi Ito, Christoph Reisinger, and Yufei Zhang, *A neural network based policy iteration algorithm with global $H^2$-superlinear convergence for stochastic games on domains*, Foundations of Computational Mathematics, 21 (2021), pp. 331–374.

[19] Christoph Reisinger and Yufei Zhang, *A penalty scheme and policy iteration for nonlocal HJB variational inequalities with monotone drivers*, Computers and Mathematics with Applications, 93 (2021), pp. 199–213.

[20] Roxana Dumitrescu, Christoph Reisinger, and Yufei Zhang, *Approximation schemes for mixed optimal stopping and control problems with nonlinear expectations and jumps*, Applied Mathematics & Optimization, 83 (2021), pp. 1387–1429.

[21] Christoph Reisinger and Yufei Zhang, *Rectified deep neural networks overcome the curse of dimensionality for nonsmooth value functions in zero-sum games of nonlinear stiff systems*, Analysis and Applications, 18 (2020), pp. 951–999.

[22] Christoph Reisinger and Yufei Zhang, *Error estimates of penalty schemes for quasi-variational inequalities arising from impulse control problems*, SIAM Journal on Control and Optimization, 58 (2020), pp. 243–276.

[23] Christoph Reisinger and Yufei Zhang, *A penalty scheme for monotone systems with interconnected obstacles: convergence and error estimates*, SIAM Journal of Numerical Analysis, 57 (2019), pp. 1625–1648.

## Conference publications

[1] Philipp Plank, Fernando Acero, and Yufei Zhang, *Reinforcement Learning with Hawkes-Structured Networks for Market Making*, 7th ACM International Conference on AI in Finance, forthcoming, 2026. [Preprint version.]

[2] Xuan Di, Anran Hu, Zhexin Wang, and Yufei Zhang, *$\alpha$-Potential Games for Decentralized Control of Connected and Automated Vehicles*, Proceedings of 23rd IFAC World Congress, forthcoming, 2026 [Preprint version.]

[3] Xinshi Chen, Yufei Zhang, Christoph Reisinger, and Le Song, *Understanding Deep Architectures with Reasoning Layer*, Advances in Neural Information Processing Systems (NeurIPS 2020), 33 (2020), pp. 1240–1252.

## Preprints

[1] Galit Ashkenazi-Golan, Edward Plumb, Clemens Possnig, Yufei Zhang, *Delegate Pricing to Algorithms: When Slow and Steady Wins the Race*, preprint, arXiv:2610.11675, 2026.

[2] Dantong Chu, Xuefeng Gao, Yufei Zhang, *Learning under Opponent Unawareness in Linear-Quadratic Stochastic Games*, preprint, arXiv:2608.08268, 2026.

[3] Anran Hu, Zhexin Wang, Yufei Zhang, Xuan Di, *Decentralized Decision-Making among Heterogeneous Autonomous Vehicles: An $\alpha$-Potential Game Framework*, preprint, arXiv:2609.38731, 2026.

[4] Ziheng Cheng, Xin Guo, Huyên Pham, and Yufei Zhang, *Actor-Critic Learning for Extended Mean Field Control with Deterministic Policies*, preprint, arXiv:2607.11005, 2026

[5] Viet Vu, Renyuan Xu, Jiacheng Zhang, and Yufei Zhang, *Expressivity and Statistical Trade-offs in Diffusion Policy Learning*, preprint, arXiv:2607.07967, 2026

[6] Jialun Cao, Fernando Acero, David Šiška, and Yufei Zhang, *Entropy Regularization Improves Policy Robustness in Continuous-Time Reinforcement Learning*, preprint, arXiv:2607.03168, 2026

[7] Xin Guo, Meng Wang, and Yufei Zhang, *Limit Theory for $N$-Player $\alpha$-Potential Games*, preprint, arXiv:2606.09815, 2026

[8] Philipp Plank and Yufei Zhang, *Learning Distributed Equilibria in Linear-Quadratic Stochastic Differential Games: An $\alpha$-Potential Approach*, Minor Revision at Mathematics of Operations Research, arXiv: 2602.16555, 2026.

[9] Ziheng Cheng, Xin Guo, and Yufei Zhang, *Deterministic Policy Gradient for Reinforcement Learning with Continuous Time and State*, preprint, arXiv:2509.23711, 2025

[10] Christoph Reisinger, Wolfgang Stockinger, Maria Olympia Tsianni, and Yufei Zhang, *Convergence Rates of Time Discretization in Extended Mean Field Control*, Minor Revision at IMA Journal of Numerical Analysis, arXiv:2509.00904, 2025.

[11] Xin Guo, Xinyu Li, and Yufei Zhang, *Distributed games with jumps: An $\alpha$-potential game approach*, Minor Revision at Mathematical Finance, arXiv: 2508.0192, 2025.

[12] Philipp Plank, Yufei Zhang, *Policy optimization for continuous-time linear-quadratic graphon mean field games*, Minor Revision at Journal of Machine Learning Research, arXiv: 2506.05894, 2025.

[13] Matthieu Meunier, Christoph Reisinger, and Yufei Zhang, *Efficient learning for entropy-regularized Markov Decision Processes via Multilevel Monte Carlo*, arXiv: 2503.21224, 2025.

[14] Xin Guo, Anran Hu, Jiacheng Zhang, Yufei Zhang, *Continuous-time mean field games: a primal-dual characterization*, arXiv:2503.01042, 2025.

[15] Eyal Neuman, Wolfgang Stockinger, and Yufei Zhang, *An offline learning approach to propagator models*, Revision at Mathematical Finance, arXiv:2309.02994, 2023.

PH.D SUPERVISION

- Nicolo Toia (2026-now, Imperial College London).
- Jack Daly (2026-now, Imperial College London).
- Meng Wang (2025-now, Imperial College London).
- Philipp Plank (2024-now, Imperial College London).

CONFERENCES ORGANISATION

- Senior Program Committee, 7th ACM International Conference on AI in Finance (ICAIF 2026), Milan (Nov. 14-17 2026).
- Senior Program Committee, 6th ACM International Conference on AI in Finance (ICAIF 2025), Singapore (Nov. 15-18 2025).
- Corresponding Organiser, Isaac Newton Institute Satellite Program “Bridging Stochastic Control and Reinforcement Learning", Cambridge (Nov. 3-28, 2025)
- Organizing Committee, 9th-London-Paris Bachelier Workshop, London (Nov. 2025).
- Organizing Committee, BIRS workshop “Advances in Stochastic Control and Reinforcement Learning: Theory and Application", Banff, Canada (April 27-May 2, 2025)
- Organizing Committee, Imperial-LSE Workshop on Financial Mathematics, London (May 2025)
- Scientific Committee, “London–Oxford–Warwick Mathematical Finance Workshop", Oxford (Jan. 9-10, 2025).
- Organizing Committee, 8th-London-Paris Bachelier Workshop, Paris (September 2024).
- Organizing Committee, ETH–Hong Kong–Imperial Mathematical Finance Workshop, London (June 2024).
- Organizing Committee, 7th-London-Paris Bachelier Workshop, London (September 2023).

ADMINISTRATIVE DUTIES

- External Examiner to M.Sc. in Mathematical and Computational Finance, University of Oxford (2026-now).
- Member of the Research Committee of the Department of Mathematics, Imperial College (2024-now).
- Organiser of the Finance and Stochastics seminar at Imperial College (2023-2025).

PROFESSIONAL SERVICE

**Editorial Service**

- Frontiers of Mathematical Finance, Associate Editor (2026âĂŞ)
- Mathematics of Operations Research, Associate Editor (2026âĂŞ)

**Referee Service**

- in **mathematical finance**: Applied Mathematical Finance, Finance and Stochastics, Journal of Computational Finance, Market Microstructure and Liquidity, Mathematical Finance, SIAM Journal on Financial Mathematics, and others.
- in **machine learning**: Journal of Machine Learning Research, Journal of Machine Learning, Advances in Neural Information Processing Systems (NeurIPS 2021), Conference on Mathematical and Scientific Machine Learning (MSML 2020), and others.
- in **control and optimization**: Applied Mathematics and Optimization, Automatica, IEEE Transactions on Automatic Control, Operations Research, SIAM Journal on Control and Optimization, and others.
- in **probability and statistics**: Annals of Applied Probability, Annals of Statistics, Stochastic Processes and Their Applications, and others.
- in **computational mathematics**: Advances in Computational Mathematics, SIAM Journal on Scientific Computing, and others.
- in **other areas of applied mathematics**: Discrete and Continuous Dynamical Systems Series B, Journal of Mathematical Analysis and Applications, and others.

PLENARY TALKS

- *International Conference on Computational Finance*, Oxford, Sept. 1-4, 2026.
- *Byrne Workshop on Stochastic Analysis in Finance and Insurance*, Ann Arbor, June 2-4, 2026.
- *Quantifying and Managing Uncertainty (QUAMU) Focus Group*, Bielefeld, Mar. 2-13, 2026.
- *Advances in Financial Mathematics 2026*, Paris, Jan. 27-30, 2026.

INVITED TALKS

- *Stochastic Control and Data-Driven Learning*, Berlin, Oct. 21, 2026.
- *Conference in honour of Dr Martin Clark*, London, May 25-26, 2026
- *Berlin Probability Colloquium*, Berlin, May 20, 2026
- *Oxford-PKU Joint Conference on Quantitative Finance and Data Science*, Beijing, April 20-23, 2026.
- *Rome Joint Seminars on Stochastic Dynamics Control and Learning*, Rome, Mar. 26, 2026.
- *Workshop on Stochastic Control, Financial Technology, and Machine Learning*, Hong Kong, Dec. 11-14, 2025.
- *Columbia-NYU Financial Engineering Colloquium*, New York, Oct. 9, 2025.
- *Probability/Math Finance Seminar*, Carnegie Mellon University, Oct. 6, 2025.
- *Advances in Mathematics of Randomness for Handling Risks in Finance and Insurance*, Luminy, Sept. 15-19, 2025.
- *Stochastics & Computational Finance 2025*, Lisbon, Sept. 2-5, 2025.
- *CRiSM 2.0 Conference*, University of Warwick, May 21-23, 2025.
- *2nd ETH-HKG-ICL Mathematical Finance Workshop*, Hong Kong, April 22-25, 2025.
- *ISOR Colloquium*, University of Vienna, Vienna, Austria, Mar. 31, 2025.
- *Statistics Seminar Series*, Collegio Carlo Alberto, Torino, Italy, Feb. 12-14, 2025.
- *Modeling, Learning and Understanding: Modern Challenges between Financial Mathematics, Financial Technology and Financial Economics*, Banff, Nov. 10-15, 2024.

- *12th Bachelier World Congress of the Bachelier Finance Society*, Rio de Janeiro, July 8-12, 2024.
- *New Trends and Challenges in Stochastic Differential Games*, Banff, June 23-28, 2024.
- *ETH-Hong Kong-Imperial Mathematical Finance Workshop*, London, June 17-20, 2024.
- *Probability for Machine Learning seminar*, Oxford, June 12, 2024.
- *Mathematical Finance seminar*, Bielefeld, June 5, 2024.
- *Bachelier Seminar*, Paris, April 17, 2024.
- *Fields-CFI Bootcamp on Machine Learning in Quantitative Finance*, Toronto, April 25-26, 2024.
- *Recent Advances in Stochastic Control, Machine Learning and Quantitative Finance*, Shanghai, April 15-19, 2024.
- *IMSI Workshop on Decision Making and Uncertainty*, Chicago, Feb. 2-9, 2024.
- *CityU-NUS MFG/MFC seminar*, Jan. 30, 2024.
- *16th International Conference of the ERCIM WG on Computational and Methodological Statistics*, Berlin, Dec. 16-18, 2023.
- *7th London-Paris Bachelier Workshop on Mathematical Finance*, London, Sept. 18-19, 2023.
- *The Second HKSIAM Biennial Meeting*, Hong Kong, Aug. 28-Sept. 1, 2023.
- *Recent Advances on Quantitative Finance*, Hong Kong, Aug. 27-30, 2023
- *10th International Congress on Industrial and Applied Mathematics*, Tokyo, Aug. 20-25, 2023.
- *11th Advanced Mathematical Methods for Finance Conference*, Bielefeld, June 26-30, 2023.
- *Stochastic Analysis and Math Finance Seminar*, Berlin, June 22, 2023.
- *Berlin Probability Colloquium*, Berlin, June 21, 2023.
- *North British Probability Seminar*, The University of Edinburgh, June 14, 2023.
- *Data Science Seminar*, The University of Essex, May 11, 2023.
- *2nd Workshop on Machine Learning for PDEs*, Imperial College London, Apr. 3-4, 2023.
- *Probability Seminar*, The University of Bath, Jan. 9, 2023.
- *World Online Seminars on Machine Learning in Finance*, Virtual, Nov. 22, 2022.
- *Machine Learning and Optimal Control*, Royal Statistical Society, Virtual, Oct. 19, 2022.
- *Finance and Stochastic Seminar*, The University of Sydney, Oct. 11, 2022.
- *London-Paris Bachelier Workshop on Mathematical Finance*, Paris, France, Sept. 15-16, 2022.
- *Machine Learning for PDEs*, London, UK, Sept. 6-8, 2022.
- *The 9th International Colloquium on BSDEs and Mean Field Systems*, Annecy, France, June 26–July 1, 2022.
- *IMSI Workshop on Machine Learning and Mean-Field Games*, Chicago, May 23–27, 2022.
- *Maxwell Institute Probability Seminar*, Heriot-Watt University and University of Edinburgh, Mar. 24, 2022.
- *Finance and Stochastic Seminar*, Imperial College London, Mar. 23, 2022.
- *Financial/Actuarial Mathematics Seminar*, University of Michigan, Virtual, Mar. 16, 2022.
- *SIAG/FME virtual seminar*, Virtual, Mar. 10, 2022.
- *15th German Probability and Statistics Days*, Virtual, Sept. 27-Oct. 1, 2021.
- *2nd Fudan-Warwick Workshop on Financial Mathematics and Stochastic Analysis*, University of Warwick, UK, July 30–31, 2019.
- *3rd International Conference on Computational Finance*, A Coruña, Spain, July 8–12, 2019.
- *International Workshop on PDE-Constrained Optimization, Optimal Controls and Applications*, Sanya, China, Dec. 10–14, 2018.
- *10th Oxford-Berlin Young Researchers Meeting on Applied Stochastic Analysis*, Oxford, United Kingdom, Nov. 29–Dec. 1, 2018.
- *14th Viennese Conference on Optimal Control and Dynamic Games*, Vienna, Austria, July 3–6, 2018.

## TEACHING EXPERIENCE

- Lecturer at Imperial College London:

- Summer School on Systematic Trading — July 2025
- Simulation Methods for Finance — Spring 2024, 2025
- Advances in Machine Learning — Spring 2025
- Interest Rate Models — Spring 2025
- Summer School on Introduction to Modern Machine Learning — July 2024

- Lecturer at London School of Economics:
  - Stochastic Process — Fall 2021, 2022
  - Stochastic Simulation — Spring 2023
  - Computational Methods in Finance and Insurance — Spring 2022, 2023

- Tutor at University of Oxford:
  - Analysis II — Spring 2021
  - Fixed Income — Spring 2021
  - Financial Derivatives — Fall 2020
  - Introduction to Probability — Fall 2020
  - Advanced Numerical Methods — Spring 2020
  - Numerical Methods — Fall 2019

## Professional Memberships

- Member, Bachelier Finance Society
- Member, Society for Industrial and Applied Mathematics

*Last updated on November 4, 2026*